

CC Lab

1. Installing Python on Windows

Option 1: Microsoft Store (Recommended for Beginners)

1. Open the **Microsoft Store** from your Start Menu.
2. Search for **Python**.
3. Select the latest version (e.g., **Python 3.12**).
4. Click **Get** or **Install**.
5. Wait for installation to complete.

Option 2: Using `winget` (Command Line)

1. Press `Win + X` and select **Terminal** or **Command Prompt**.
2. Type the following command and press Enter:

```
winget install Python.Python.3
```

3. Wait for installation to finish.

Option 3: Download from Browser

1. Go to <https://www.python.org/downloads/>
2. Click **Download Python 3.x.x** (latest version).
3. Run the downloaded installer.
4. **Important:** Check the box **"Add Python to PATH"** before clicking **Install Now**.

2. Verify Python Installation

Open **Command Prompt** or **Terminal** and type:

```
python --version
```

You should see something like `Python 3.12.0`.

3. Running Your Python Scripts

Step 1: Open Terminal in Your Project Folder

- Open **File Explorer** and navigate to your project folder:
In my case: `D:\UET\6th semester\CC Lab\lab 01`
- In the address bar, type `cmd` and press Enter.
This opens Command Prompt in the correct folder.

Step 2: Run the Scripts

For Task 1 (`task_01.py`):

```
python task_01.py
```

- This will print the tokens for each test expression.

For Task 3 (`task_03.py`):

```
python task_03.py
```

- This will print tokens, and for expressions, the result or variable assignment.

4. What Each Script Does

task_01.py

- **Purpose:** Tokenizes mathematical expressions (breaks them into numbers, operators, variables, etc.).
- **Output:** Prints each token for several sample expressions.

task_03.py

- **Purpose:**
 - Tokenizes, parses, and evaluates mathematical expressions.
 - Supports variable assignment (e.g., `x = 10`).
 - Handles operator precedence and parentheses.
- **Output:**
 - Prints tokens for each expression.
 - Prints the result for calculations.
 - Prints variable assignments.

5. Sample Output

```
Expression: 3 + 4 * 2
Tokens: ['NUMBER:3', 'PLUS', 'NUMBER:4', 'MULTIPLY', 'NUMBER:2']
Result: 11

Expression: (3 + 4) * 2
Tokens: ['LPAREN', 'NUMBER:3', 'PLUS', 'NUMBER:4', 'RPAREN', 'MULTIPLY', 'NUMBER:2']
Result: 14

Expression: x = 10
Tokens: ['IDENTIFIER:x', 'ASSIGN', 'NUMBER:10']
Variable x assigned value 10

Expression: y = 5
Tokens: ['IDENTIFIER:y', 'ASSIGN', 'NUMBER:5']
Variable y assigned value 5

Expression: x + y * 2
Tokens: ['IDENTIFIER:x', 'PLUS', 'IDENTIFIER:y', 'MULTIPLY', 'NUMBER:2']
Result: 20

Expression: ((3 + 5) * 2) - 4 / 2
Tokens: ['LPAREN', 'LPAREN', 'NUMBER:3', 'PLUS', 'NUMBER:5', 'RPAREN', 'MULTIPLY', 'NUMBER:2', 'RPAREN', 'MINUS', 'NUMBER:4', 'DIVID
Result: 14
```

6. Troubleshooting

- If you get 'python' is not recognized, Python is not installed or not added to PATH. Reinstall and check **"Add Python to PATH"**.
- If you get a `ModuleNotFoundError`, make sure you are in the correct folder and the file exists.